

Do Markets Know Inflation First?

Evaluating Kalshi's KXCPI prediction markets as real-time forecasts of U.S. CPI

1

THE QUESTION

Can a betting market
forecast inflation?

Every month, real money bets on the CPI print

1

Prices, not opinions

Traders buy and sell contracts on where CPI will land. Prices move continuously and encode a probability-weighted consensus — updated by anyone with money on the line.

57

monthly CPI releases
Nov 2021 – Jul 2026

2

The efficient-market bet

If markets aggregate information efficiently, those prices could rival professional forecasters — with no econometric model, just incentives to be right.

3,484

contract-days of
live market quotes

3

What we tested

We put that to the test on Kalshi's KXCPI threshold contracts across every monthly release since the market opened.

\$0

cost of the forecast —
the market is free

How the market works: CPI threshold contracts

Contract	Price = P(above)	Implies outcome	Prob.
CPI > 0.0%	95¢	≤ 0.0%	5%
CPI > 0.1%	78¢	= 0.1%	17%
CPI > 0.2%	35¢	= 0.2%	43%
CPI > 0.3%	8¢	= 0.3%	27%
		> 0.3%	8%

Differencing adjacent threshold prices turns a stack of yes/no contracts into a full probability distribution over the CPI print.

THE MARKET'S POINT FORECAST

Implied mean = $\Sigma (\text{probability} \times \text{outcome})$

$$.05(0.0) + .17(0.1) + .43(0.2) + .27(0.3) + .08(0.4)$$

$$= 0.22\%$$

one number the market is willing to back

2

THE APPROACH

**From prices to a
fair comparison**

Methodology: handling the raw data from Kalshi

Three handling rules — each demonstrated on real KXCPI data in the appendix

1

Price = live bid/ask midpoint

A last-trade price often does not exist, but live quotes always do. We price every contract at the bid/ask midpoint: bid 25¢, ask 35¢ → a 30% probability.

53%

of quoted contract-days
have zero trades

2

Distribution sums to one by construction

Subtracting each threshold price from the one below slices the whole 100% into outcomes — the total is exact by construction, nothing to rescale.

1.000

probability total on
all 3,484 event-days

3

Monotonicity: measure, fix, report

$P(\text{CPI} > 0.2\%)$ can never exceed $P(\text{CPI} > 0.1\%)$; thin quotes sometimes break this rule. A running-minimum cap repairs it, and every fix is logged.

12%

of days need a fix —
median under 1¢

Benchmarking: Kalshi vs. the Cleveland Fed nowcast

1

The benchmark: a professional model

The Cleveland Fed inflation nowcast: a statistical model that predicts CPI, published every business day. We match it to all 57 settled Kalshi months.

57

Kalshi months matched
to the daily nowcast

2

A fair comparison

Both are graded against the same answer key — the first-release CPI print — read on the same date: 1, 7, 14, and 30 days before each release. No one sees the future.

4

checkpoints per month:
1, 7, 14, 30 days out

3

A statistical test, not eyeballing

The Diebold–Mariano test asks: is one forecaster genuinely more accurate, or just luckier month to month? The verdict comes in the results section.

56

shared months in the
significance test

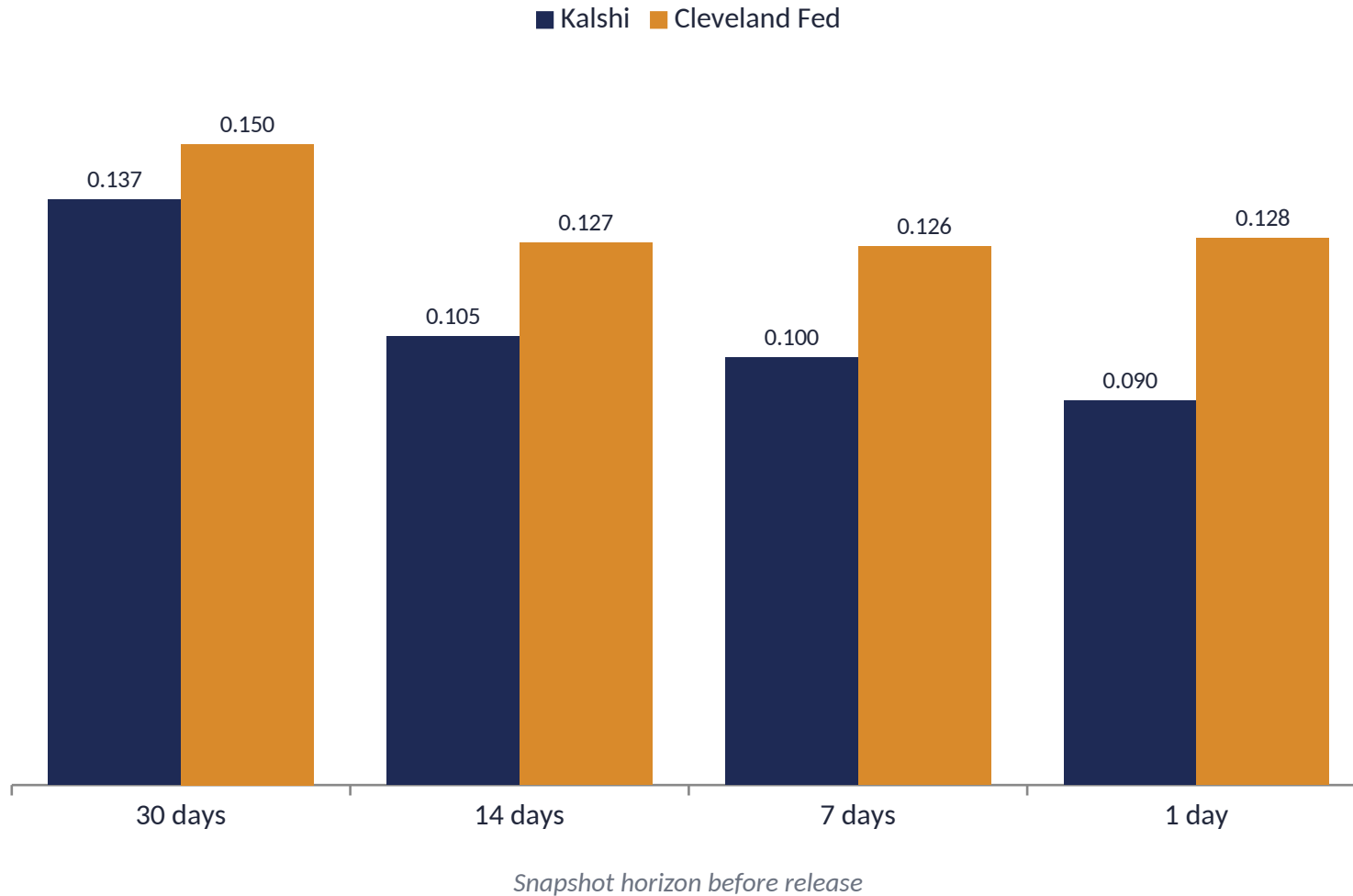
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KEY FINDINGS

**The market beats
the model**

The market beats the model at every horizon

Mean absolute error vs. the Cleveland Fed nowcast • 57 settled releases, real-time vintages



READING IT

Kalshi is lower — more accurate — at all four horizons.

Its edge widens as the release nears:
0.090 vs 0.128 pp one day out.

Holds under RMSE too — 0.125 vs 0.166
one day out — which weights the big-surprise months most.

Is the gap real? The Diebold–Mariano test

Paired test of equal accuracy on $|\text{error}| \cdot \text{loss} \cdot n = 56$ common events

Horizon	Mean loss diff	t-stat	p-value	Verdict
1 day	-0.037	-3.26	0.001	significant
7 days	-0.024	-2.06	0.040	significant
14 days	-0.020	-1.62	0.104	suggestive
30 days	-0.010	-0.61	0.542	inconclusive

Negative loss difference $\rightarrow$ Kalshi's errors are smaller. $p < 0.05 \rightarrow$ the gap is unlikely to be luck.

VERDICT

Kalshi's advantage is statistically significant in the final week

$p = 0.001$ at one day, 0.04 at seven. The direction favors Kalshi at every horizon; 56 events limit the test's power further out.

4

TAKEAWAYS

What it means,
and what's next

What holds it back — stated honestly

- 1 Short sample**

57 months, dominated by the 2021–23 inflation spike. Planning a pre/post-2023 split to check the edge holds in calmer regimes.
- 2 Thin liquidity**

Many zero-volume days and wide spreads. The implied mean occasionally leaves the $[0,1]$ range — a symptom we measure and flag.
- 3 A missing month**

October 2025 CPI was never released (government shutdown). We skip it explicitly rather than impute a value.

What we learned — and what comes next

1

The market beats the model

One day before release the market misses by 0.090pp on average vs 0.128 for the Fed nowcast — closer in 33 of 57 months, and too consistent to be luck ($p = 0.001$).

33 of 57

months the market was
closer to the print

2

The edge grows as data arrives

Kalshi improves from a 0.137pp miss 30 days out to 0.090 the day before; the nowcast plateaus near 0.13. Both are unbiased — the win is tighter misses, not direction.

-34%

error shrinkage from
30 days out to release eve

3

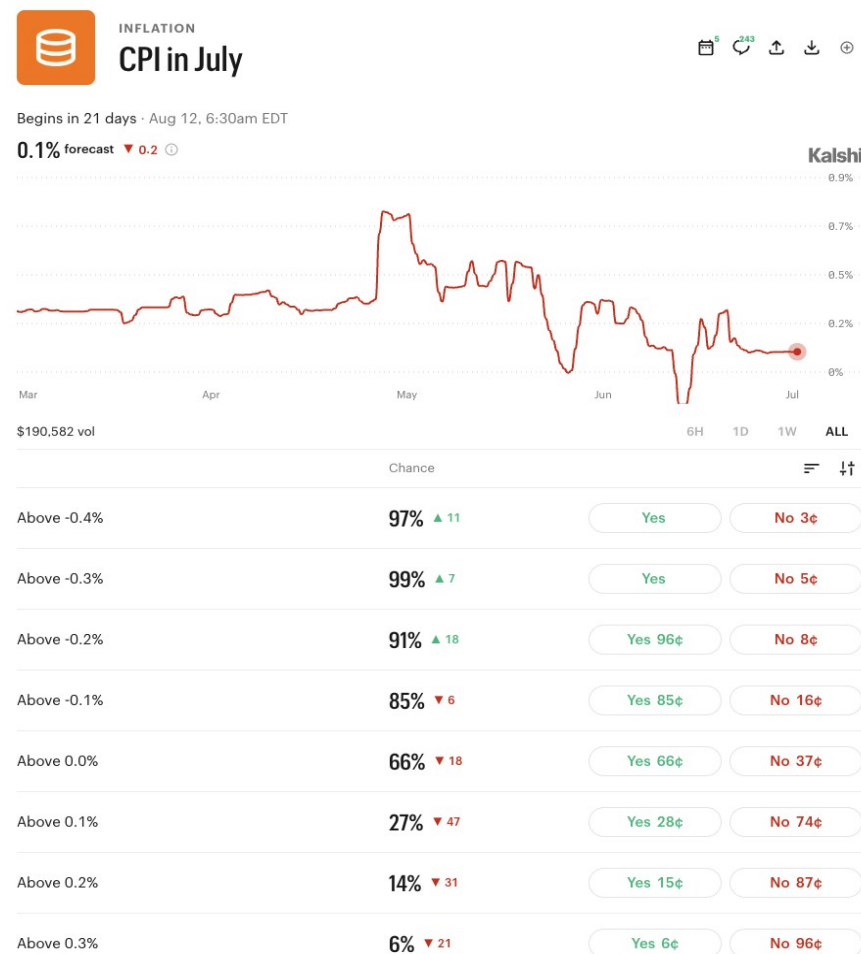
What comes next

Add the economist consensus median as a second benchmark, stress-test the liquidity filter and tail assumptions, and extend to year-over-year CPI (KXCPIYOY).

Appendix

Technical backup — worked examples on real KXCPI days, for Q&A

The market: monthly CPI threshold contracts on Kalshi



Example: prices → probability distribution

Contract	Price = P(above)	⇒ Outcome	Prob.
CPI > 0.0%	95¢	≤ 0.0%	5%
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Implied mean — the market's point forecast

$\Sigma \text{probability} \times \text{outcome}$: $.05 \times 0.0 + .17 \times 0.1 + .43 \times 0.2 + .27 \times 0.3 + .08 \times 0.4 = 0.216\%$

Rule 1 — Price = bid/ask midpoint, not last trade

Real day: KXCPI-25JAN-T0.4 on Jan 3, 2025 — "Will Jan 2025 CPI rise more than 0.4%?"

The problem On 53% of contract-days with live quotes, nothing traded — no trade price exists, and stale prints can be days old.

The rule Use the midpoint of the closing bid and ask. Quotes are enforceable orders, so they are current even on zero-trade days — and the truth is bracketed between them.

Trades that day (volume)	0 — nothing traded
Last trade price	(none)
Best bid (sell now)	\$0.25
Best ask (buy now)	\$0.35
Midpoint → P(CPI > 0.4%)	(0.25+0.35)/2 = 30%

A last-trade rule would have no price here; the midpoint gives a live 30% forecast. (January CPI printed 0.5% — the contract settled YES.)

Rule 2 — Probabilities sum to 1 by construction

Real day: CPI-23AUG on Sep 11, 2023 (2 days pre-release) — all four informative thresholds

Outcome slice	Formula	Value
CPI ≤ 0.4%	$1 - S(0.4)$	$1 - 0.985 = 0.015$
CPI = 0.5%	$S(0.4) - S(0.5)$	$0.985 - 0.820 = 0.165$
CPI = 0.6%	$S(0.5) - S(0.6)$	$0.820 - 0.515 = 0.305$
CPI = 0.7%	$S(0.6) - S(0.7)$	$0.515 - 0.100 = 0.415$
CPI > 0.7%	$S(0.7)$	0.100
Total	everything cancels	= 1.000 exactly

$$(1-S_1) + (S_1-S_2) + (S_2-S_3) + (S_3-S_4) + S_4 = 1$$

A telescoping sum — every price cancels, only the 1 survives, for any prices on any day. Verified at exactly 1.0 on all 3,484 event-days, so incoherence must show up as Rule 3.

Two days later the BLS printed 0.6% — the bolded slice.

Rule 3 — Out-of-order prices: measure, fix, report

Real day: CPI-24JAN on Feb 1, 2024 (12 days pre-release) — two prices sit above the one below

Threshold	Quoted P(above)	After fix	Slice
> -0.2%	0.900	0.900	0.000
> -0.1%	0.970 <i>x</i>	0.900	0.000
> 0.0%	0.905 <i>x</i>	0.900	0.380
> 0.1%	0.520	0.520	0.455
> 0.2%	0.065	0.065	0.065

The problem $P(>-0.1\%)$ can't exceed $P(>-0.2\%)$ — but they trade independently and stale quotes invert them. Differencing would give a negative probability.

The fix Running minimum — cap each price at the one below it. Both offenders drop to 0.900; the recorded `mono_violation` = 0.07.

The accounting Violation size is logged every day: nonzero on 12%, median < 1¢, max 41¢, correlated 0.34 with spreads — a thin-market symptom, reported not hidden.

=== ACCURACY BY SOURCE x HORIZON ===

	source	horizon	n	MAE	RMSE	bias
	kalshi	1	57	0.0901	0.1245	0.0018
	cleveland_fed	1	56	0.1282	0.1655	0.0101
	kalshi	7	57	0.1004	0.1372	0.0017
	cleveland_fed	7	56	0.1261	0.1637	0.0114
	kalshi	14	57	0.1054	0.1436	-0.0012
	cleveland_fed	14	56	0.1273	0.1657	0.0090
	kalshi	30	55	0.1374	0.1850	0.0047
	cleveland_fed	30	54	0.1496	0.1983	0.0177

=== Diebold-Mariano: kalshi vs cleveland_fed (|error| loss) ===

horizon	n	mean_loss_diff	t_stat	p_value	winner
1	56	-0.0367	-3.26	0.001	kalshi
7	56	-0.0241	-2.06	0.040	kalshi
14	56	-0.0202	-1.62	0.104	kalshi
30	54	-0.0100	-0.61	0.542	kalshi